

**ABSOLUTE HODGE CYCLES AND PERIODS OF
INTEGRALS ON ABELIAN VARIETIES**

by

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We show here how a suitable theory of “motives” makes it possible to recover certain results of Shimura on the periods of abelian varieties of CM type [5]. It seemed instructive to begin with an informal description of the properties of motives and of their realizations; this is done in Section 1. Section 2 interprets certain results of Shimura by means of motives. Section 3 defines absolute Hodge cycles and the theory of motives subordinate to them. Section 4 gives an upper bound for the transcendence degree of the field generated by the periods of an arbitrary abelian variety.

§ 1. THE LANGUAGE OF MOTIVES

Let k be a field of characteristic 0. There are several definitions of “motives over k ”, which are not known to be equivalent, each having its own

virtues. In Section 3 we shall specify which one is useful to us.

The motives over k form an additive category equipped with a tensor product. Every smooth projective variety X over k defines a motive $h(X)$ over k , the motivic cohomology of X , and every motive over k is obtained from a motive of this form (by cutting it into pieces and by Tate twisting). Finally, the various cohomology theories for algebraic varieties give rise to “realization” functors for motives.

Every abelian variety A over k defines a motive $T(A)$ over k , its motivic H_1 , and the functor $A \mapsto T(A)$ is fully faithful. The advantage of embedding the category of abelian varieties in that of motives is that one can take tensor products. But the objects obtained are not geometric: there are no “functions” on a motive.

The rank of a motive is the dimension of any one of its realizations, which we now describe.

If $k = \mathbf{C}$, the Betti realization of a motive M is a free abelian group of finite rank $H_B(M)$, endowed with a Hodge structure, that is, with a decomposition

$$H_B(M) \otimes_{\mathbf{Z}} \mathbf{C} = \bigoplus_{p+q=n} H^{p,q} \quad (\text{with } \overline{H^{p,q}} = H^{q,p}).$$

Let us illustrate this with the motive $T(A)$, which is the H_1 of an abelian variety A . We define $H_B(T(A)) = H_1(A, \mathbf{Z})$, the homology of $A = A(\mathbf{C})$ viewed as a topological space.

Let $\text{Lie}(A)$ be the Lie algebra of the analytic group A , and let $\int : H_1(A, \mathbf{Z}) \rightarrow \text{Lie}(A)$ be the homomorphism “integration over a 1-cycle”. There is an exact sequence of complex analytic groups

$$0 \longrightarrow H_1(A, \mathbf{Z}) \xrightarrow{\int} \text{Lie}(A) \xrightarrow{\exp} A \longrightarrow 0.$$

The Hodge structure on $H_1(A, \mathbf{C})$ is such that $H_1(A, \mathbf{C}) = H^{0,-1} \oplus H^{-1,0}$. One obtains $H^{0,-1}$ as the kernel of the map $H_1(A, \mathbf{C}) \rightarrow \text{Lie}(A)$, the $\mathbf{C}$ -linear extension of

$$\int : H_1(A, \mathbf{Z}) \rightarrow \text{Lie}(A).$$

We now work over a field of characteristic 0 and choose an algebraic closure $\bar{k}$ of k . For every prime number ℓ , the ℓ -adic realization of a motive M is a free $\mathbf{Z}_\ell$ -module of finite type $H_\ell(M)$ (also endowed with an action of the Galois group of $\bar{k}$ over k). When $k = \mathbf{C}$, there is an isomorphism

$$H_\ell(M) \simeq H_B(M) \otimes \mathbf{Z}_\ell.$$

If A is an abelian variety over k and $\bar{k}$ is an algebraic closure of k , the ℓ -adic realization of $T(A)$ is denoted $T_\ell(A)$; it is the $\mathbf{Z}_\ell$ -module

$$T_\ell(A) = \varprojlim_n A_{\ell^n}(\bar{k}) \quad (\text{where } A_{\ell^n}(\bar{k}) \text{ is the group of points of order } \ell^n \text{ in } A(\bar{k})).$$

If $k = \mathbf{C}$, the isomorphism $T_\ell(A) \simeq H_B(T(A)) \otimes \mathbf{Z}_\ell$ is easy to describe. For every n , the projection of $\gamma \otimes x \in H_B(T(A)) \otimes \mathbf{Z}_\ell$ to $A_{\ell^n}(\mathbf{C})$ is computed as follows: if $m \in H_B(T(A))$ is such that $x - m \in \ell^n(H_B(T(A)) \otimes \mathbf{Z}_\ell)$, the exponential of $m/\ell^n \in H_1(A, \mathbf{Q})$ is the required point of order ℓ^n .

The de Rham realization of M is a finite-dimensional vector space $H_{DR}(M)$ over k , equipped with a decreasing filtration $F^\bullet$ (the Hodge filtration). When $k = \mathbf{C}$, there is an isomorphism

$$H_B(M) \otimes \mathbf{C} \simeq H_{DR}(M).$$

Under this isomorphism, the Hodge filtration is given by $F^p = \bigoplus_{p' \geq p} H^{p', q}$. In the case of the motive $T(A)$, recall [7] the existence of a “universal extension” E of A by a vector group, fitting into an exact sequence of algebraic groups

$$0 \longrightarrow H^1(A, \mathcal{O}_A)^* \longrightarrow E \longrightarrow A \longrightarrow 0.$$

We have $H_{DR}(T(A)) = \text{Lie}(E)$, and the Hodge filtration is

$$0 = F^1 \subset F^0 = H^1(A, \mathcal{O}_A)^* \subset F^{-1} = \text{Lie}(E).$$

If $k = \mathbf{C}$, then $H_1(E, \mathbf{Z}) \xrightarrow{\sim} H_1(A, \mathbf{Z})$, giving a morphism

$$H_1(A, \mathbf{C}) \xleftarrow{\sim} H_1(E, \mathbf{C}) \xrightarrow{\int} \text{Lie}(E) = H_{DR}(T(A)).$$

This is the desired isomorphism.

Examples: **1)** Here $k = \mathbf{Q}$. We shall define the Tate motive $\mathbf{Z}(1)$; it is the H_1 of the algebraic group $\mathbf{G}_m$. To obtain its various realizations, substitute $\mathbf{G}_m$ for A in the definition of $T(A)$. This gives

- $H_B(\mathbf{Z}(1)) = 2\pi i \cdot \mathbf{Z}$, the kernel of $\exp : \mathbf{C} \rightarrow \mathbf{C}^\times$;
- $H_B(\mathbf{Z}(1)) \otimes \mathbf{C} = H^{-1,-1}$;
- $H_\ell(\mathbf{Z}(1)) = \mathbf{Z}_\ell(1) = T_\ell(\mathbf{G}_m) = \varprojlim_n \mu_{\ell^n}(\bar{\mathbf{Q}})$;
- $H_{DR}(\mathbf{Z}(1)) = \mathbf{Q}$.

The isomorphism $H_B(\mathbf{Z}(1)) \otimes \mathbf{C} \xrightarrow{\sim} H_{DR}(\mathbf{Z}(1)) \otimes \mathbf{C}$ sends $(2\pi i) \otimes 1$ to $2\pi i$. Recall the isomorphisms $H_1(\mathbf{G}_m) \simeq H_1(\mathbf{P}^1 - \{0, \infty\}) \simeq H_2(\mathbf{P}^1)$. Moreover, for every smooth, proper, connected algebraic curve over $\mathbf{Q}$, $H_2(X)$ is isomorphic to $H_2(\mathbf{P}^1)$ (and hence to $\mathbf{Z}(1)$). For every integer d , define a motive $\mathbf{Z}(d)$ so that $\mathbf{Z}(d) = \mathbf{Z}(d-1) \otimes \mathbf{Z}(1)$. We have $\mathbf{Z}(m+n) \simeq \mathbf{Z}(m) \otimes \mathbf{Z}(n)$ for $m, n \in \mathbf{Z}$.

2) Let E be an elliptic curve over $\mathbf{Q}$. The motive $T(E)$ has rank 2. Hence the motive $\bigwedge^2 T(E)$ has rank 1, and

$$\bigwedge^2 T(E) = \bigwedge^2 H_1(E) = H_2(E) = \mathbf{Z}(1).$$

Let us specify the isomorphism $\bigwedge^2 T(E) \simeq \mathbf{Z}(1)$ in the various realizations.

On $H_1(E(\mathbf{C}), \mathbf{R}) \simeq \text{Lie}(E)$, the complex structure defines an orientation. If (e_1, e_2) is an oriented $\mathbf{Z}$ -basis of $H_B(T(E)) = H_1(E(\mathbf{C}), \mathbf{Z})$, then $e_1 \wedge e_2$ is a basis of $H_B(\bigwedge^2 T(E))$; it is associated with $2\pi i \in H_B(\mathbf{Z}(1))$.

The isomorphism $\bigwedge^2(T_\ell(E)) \simeq \mathbf{Z}_\ell(1)$ is given by the alternating Weil pairing [6]. Finally,

$$\bigwedge^2 H_{DR}(T(E)) = \text{Hom}(H_{DR}^2(E), \mathbf{Q}).$$

Let ω and η be the usual differentials of the first and second kind. $H_{DR}^2(E)$ is generated by the cup product $\omega \wedge \eta$ of their cohomology classes. Under the isomorphism

$$H_B(\bigwedge^2 T(E)) \otimes \mathbf{C} \longrightarrow H_{DR}(\bigwedge^2 T(E)) \otimes \mathbf{C},$$

$e_1 \wedge e_2$ maps to the homomorphism sending $\omega \wedge \eta \in H_{DR}^2(E)$ to

$$\begin{vmatrix} \omega_1 & \omega_2 \\ \eta_1 & \eta_2 \end{vmatrix} \in \mathbf{C}.$$

Compatibility of the Betti and de Rham realizations of $\bigwedge^2(T(E)) \simeq \mathbf{Z}(1)$ is equivalent to the Legendre relation

$$\omega_1 \cdot \eta_2 - \omega_2 \cdot \eta_1 = 2\pi i.$$

3) Let A be an abelian variety over $\mathbf{Q}$. The cohomology class of a hyperplane section of A in a projective embedding gives a morphism of motives $\bigwedge^2 T(A) \rightarrow \mathbf{Z}(1)$. In Betti cohomology this corresponds to an alternating form $2\pi i \cdot E$ on $H_B(T(A))$ with values in $2\pi i \cdot \mathbf{Z}$. In de Rham cohomology one likewise has an alternating form E_{DR} on $H_{DR}(T(A))$ with values in $\mathbf{Q}$. It is classical that F^0 is a totally isotropic subspace. It follows that one can find differentials of the first kind $\omega_1, \dots, \omega_g$ on A and differentials of the second kind $\eta_1, \dots, \eta_g$ on A such that

$$E_{DR}(x, y) = \sum_{i=1}^g ((\omega_i, x)(\eta_i, y) - (\omega_i, y) \cdot (\eta_i, x)).$$

Compatibility between E and E_{DR} can be expressed as follows. Let $(e_1, \dots, e_{2g})$ be a basis of $H_B(T(A))$, let Ω be the $n \times 2n$ matrix with $\Omega_{ij} = \int_{e_j} \omega_i$, and let N be the $n \times 2n$ matrix with $N_{ij} = \int_{e_j} \eta_i$. Then

$${}^t N \cdot \Omega - {}^t \Omega \cdot N = 2i\pi \cdot E.$$

Finally, the ℓ -adic realization gives an alternating form on $T_\ell(A)$ with values in $\mathbf{Z}_\ell(1)$: it is the alternating Weil form associated with the divisor of the hyperplane section [6].

§ 2. MOTIVES ASSOCIATED WITH ABELIAN VARIETIES OF CM TYPE

Let k be an algebraically closed subfield of $\mathbf{C}$ – for example $\bar{\mathbf{Q}}$ – and let $\mathcal{A}$ be the subcategory of the category of motives over k generated by the $T(A)$ of abelian varieties (under tensor products, passage to duals, and direct factors). The functor “Betti realization of the motive obtained from M/k by extension of scalars to $\mathbf{C}$ ” sends $\mathcal{A}$ to the category of Hodge structures. It follows formally from Theorem 1 of Section 3 that this functor is fully faithful.

Let E be a CM field, let S be the set of these complex embeddings, and let $- : S \rightarrow S$ denote complex conjugation. To each function $\varphi : S \rightarrow \mathbf{Z}$ satisfying

$$(*) \quad \varphi(\sigma) + \varphi(\bar{\sigma}) \quad \text{is independent of } \sigma \in S,$$

associate a Hodge structure H_φ equipped with a module structure over the ring of integers $\mathcal{O}_E$ of E :

$$\begin{aligned} H_\varphi &= \mathcal{O}_E, \quad \text{with the evident module structure, and in the decomposition} \\ H_\varphi \otimes \mathbf{C} &= E \otimes \mathbf{C} \simeq \mathbf{C}^S, \quad \text{the } \mathbf{C}\text{-factor indexed by } \sigma \in S \text{ has Hodge type} \\ &\quad (\varphi(\sigma), \varphi(\bar{\sigma})). \end{aligned}$$

One verifies that the H_φ lie in the image of $\mathcal{A}$, and indeed in the image of the subcategory generated by abelian varieties of CM type. This allows one to define the motive M_φ over $\bar{\mathbf{Q}}$ as the motive whose Betti realization is H_φ .

The de Rham realization $H_{DR}(M_\varphi)$ is a vector space over $\bar{\mathbf{Q}}$ equipped with an action of E . For each $\sigma \in S$, let $H_{DR}(M_\varphi)_\sigma$ be the $\bar{\mathbf{Q}}$ -subspace on which $x \in E$ acts as $\sigma^{-1}(x) \in \bar{\mathbf{Q}}$. Each of these vector spaces is one-dimensional, generated by an element ω_σ , and $H_{DR}(M_\varphi)$ is their direct sum. Under the isomorphism

$$H_B(M_\varphi) \otimes \mathbf{C} \xrightarrow{\sim} H_{DR}(M_\varphi) \otimes \mathbf{C},$$

an element e of $E = H_B(M_\varphi) \otimes \mathbf{Q}$ corresponds to a family (e_σ) with $e_\sigma \in H_{DR}(M_\varphi)_\sigma$. It follows immediately that the complex number $p(\sigma, \varphi)$ defined by $p(\sigma, \varphi)e_\sigma = \omega_\sigma$ is independent, up to multiplication by $\bar{\mathbf{Q}}^\times$, both of e and of the choice of ω_σ . This justifies the notation.

The relation between the $p(\sigma, \varphi)$ and Shimura's invariants $p_E(\sigma, \varphi)$ (see [5]) is

$$p_E(\sigma, \varphi) = \frac{p(\sigma, \varphi)}{(2\pi i)^{\varphi(\sigma)}} \quad (\text{modulo } \bar{\mathbf{Q}}^\times),$$

which is merely a translation of the definitions.

The identities among Shimura's periods correspond to algebraic properties of the motives M_φ . Before summarizing them in a table, introduce some notation for an extension $F|E$ of CM fields. Let S_F (respectively S_E) be the set of complex embeddings of F (respectively E). For $\sigma \in S_F$, write $\sigma|E \in S_E$ for the restriction of the embedding σ to E .

For $\varphi : S_E \rightarrow \mathbf{Z}$ satisfying $(*)$, let $\text{Inf}(\varphi)$ be the function from S_F to $\mathbf{Z}$ defined by $\text{Inf}(\varphi)(\sigma) = \varphi(\sigma|E)$. This function satisfies $(*)$. For $\varphi : S_F \rightarrow \mathbf{Z}$ satisfying $(*)$, let $\text{Res}(\varphi)$ be the function from S_E to $\mathbf{Z}$ defined by

$$\text{Res}(\varphi)(\sigma) = \sum_{\tau|E=\sigma} \varphi(\tau);$$

it satisfies $(*)$.

Identities among periods	Relations among motives
1. $p(\sigma, \varphi' + \varphi'') = p(\sigma, \varphi') \cdot p(\sigma, \varphi'')$ 2. $F E$ an extension of CM fields, $\sigma \in S_F$, $\varphi : S_E \rightarrow \mathbf{Z}$ satisfying $(*)$: $p(\sigma, \text{Inf}(\varphi)) = p(\sigma E, \varphi)$ 3. $\sigma \in S_E$, $\varphi : S_F \rightarrow \mathbf{Z}$ satisfying $(*)$: $p(\sigma, \text{Res } \varphi) = \prod_{\tau E=\sigma} p(\tau, \varphi)$ 4. $p(\sigma, \underline{1}) = 2\pi i$	1. $M_{\varphi' + \varphi''} = M_{\varphi'} \otimes_E M_{\varphi''}$ 2. $M_{\text{Inf}(\varphi)} = M_{\varphi} \otimes_E F$ 3. M_{φ} is an F-motive of rank 1 and hence an E-motive of rank $d = [F : E]$ $M_{\text{Res}(\varphi)} = \bigwedge_E^d M_{\varphi}$ 4. $M_{\underline{1}} = \mathbf{Z}(-1)$ (the dual of the Tate motive)

These identities among periods recover Theorems 1 through 4 of Shimura's article [5]. Note that in [5] Shimura relates the periods $p_K(\sigma, \varphi)$ to values, at a special point, of Hilbert modular forms rational over $\bar{\mathbf{Q}}$.

§ 3. ABSOLUTE HODGE CYCLES AND MOTIVES

Let X be a smooth projective variety over $\mathbf{C}$. A Hodge cycle of codimension d on X is an element of $H^{2d}(X, \mathbf{Q})$ of type (d, d) , or equivalently an element of

$$H^{2d}(X, \mathbf{Q})(d) \stackrel{\text{def}}{=} H^{2d}(X, \mathbf{Q}) \otimes \mathbf{Z}(d)$$

of type $(0, 0)$. Observe that for $x \in H^{2d}(X, \mathbf{Q})(d)$, saying that x is of type $(0, 0)$ is equivalent to saying that x lies in F^0 . The cohomology class of an algebraic subvariety of X is a Hodge cycle. Hodge conjectured that every Hodge cycle is a linear combination

of cohomology classes of subvarieties. Since this conjecture appears inaccessible, we introduce a notion that “arithmetizes” that of a Hodge cycle.

The cycles defined here will have all the visible attributes of algebraic cycles, namely ℓ -adic and de Rham cohomology realizations. An absolute Hodge cycle of codimension d on a smooth projective variety X over a field k of characteristic 0 is a collection of cohomology classes in the groups

$$H^{2d}(X, \mathbf{Q}_\ell)(d) \quad (\text{respectively } H_{DR}^{2d}(X)(d))$$

satisfying the following two conditions:

- the element of $H_{DR}^{2d}(X)(d)$ belongs to F^0 ;
- for every embedding $k \hookrightarrow \mathbf{C}$, the different cohomology classes correspond (through the isomorphisms of Section 1) to a single rational class in the Betti cohomology of $X \otimes_k \mathbf{C}$.

There are inclusions (algebraic cycles) $\subset$ (absolute Hodge cycles) $\subset$ (Hodge cycles). We use this definition to specify the category of motives used in Sections 1 and 2. First let $\mathcal{V}(k)$ be the category of smooth projective varieties over k . One constructs an additive category $\mathcal{M}(k)$, in which the groups $\text{Hom}(M, N)$ are vector spaces over $\mathbf{Q}$, together with the following data:

- a)** a tensor product $\otimes$, associative, commutative, and distributive with respect to addition of objects;
- b)** a contravariant functor H^* from $\mathcal{V}(k)$ to $\mathcal{M}(k)$, bijective on objects, compatible with sums, and carrying products to tensor products.

In $\mathcal{M}(k)$, for connected X and Y , one has

$$\text{Hom}(H^*(X), H^*(Y)) = Z_{h.a}^{\dim(Y)}(X \times Y),$$

where $Z_{h.a}^d(Z)$ denotes the vector space over $\mathbf{Q}$ of absolute Hodge cycles of codimension d on Z . The category of motives $\mathcal{M}(k)$ is then constructed exactly as in [2], with algebraic cycles replaced by absolute Hodge cycles. By this new definition, the Künneth components of the diagonal of $X \times X$ are absolute Hodge, and in $\mathcal{M}(k)$ there is a decomposition

$$H^*(X) = \bigoplus_i H^i(X).$$

The following result was the subject of lectures at the I.H.E.S. in 1979.

Theorem 1: Every Hodge cycle on an abelian variety is absolute Hodge.

We note the following.

Corollary (Borovoi): Let A be an abelian variety over a field k of characteristic 0, and let ξ be a Hodge cycle on A . The corresponding ℓ -adic classes ξ_ℓ of ξ are fixed by an open subgroup of $\text{Gal}(\bar{k}|k)$.

§ 4. PERIODS OF INTEGRALS ON ABELIAN VARIETIES

Let A be a complex abelian variety, and let $H_1(A, \mathbf{C}) = H^{0,-1} \oplus H^{-1,0}$ be the Hodge decomposition of $H_B(A)$ (cf. Section 1). Regard $GL(H_1(A))$ as an algebraic group over $\mathbf{Q}$. There is a morphism of complex algebraic groups

$$\mathbf{G}_m \xrightarrow{\mu} GL(H_1(A, \mathbf{C}))$$

such that $\mu(z)$ acts by the homothety of ratio z on $H^{-1,0}$ and by the identity on $H^{0,-1}$. The Hodge group G of A is the Zariski closure of the subgroup $\mu(\mathbf{G}_m)$ in $GL(H_1(A))$. In other words, G is an algebraic subgroup over $\mathbf{Q}$ of $GL(H_1(A))$. A regular function on a Zariski-open subset of G that is rational over $\mathbf{Q}$ vanishes on G if and only if it vanishes on $\mu(\mathbf{G}_m)$. One checks easily that G contains the homotheties of $H_1(A)$. Here is a second, very useful description of G : consider all rational tensors of type $(-N/2, -N/2)$ in the tensor products $\bigotimes^N H_1(A)$. Then G is the subgroup of $GL(H_1(A))$ formed by those linear transformations g for which there is a scalar μ such that $g \cdot t = \mu^m t$ for every tensor t of type (m, m) . Finally, one shows that G is the group fixing all absolute Hodge cycles of type $(0, 0)$ in the mixed tensor spaces on $H_1(A)$ (mixed means covariant and contravariant).

Now suppose that A is defined over a subfield k of $\mathbf{C}$. Put $g = \dim(A)$.

Theorem 2: Let $\omega_1, \dots, \omega_g$ be differentials of the first kind on A , and let $\eta_1, \dots, \eta_g$ be differentials of the second kind on A , such that

$(\omega_1, \dots, \omega_g, \eta_1, \dots, \eta_g)$ is a basis of $H_{DR}^1(A)$. Let K be the subfield of $\mathbf{C}$ generated by k and the periods $\int_\gamma \omega_i, \int_\gamma \eta_i$ ($1 \leq i \leq g, \gamma \in H_1(A, \mathbf{Z})$); the transcendence degree of K over k is bounded above by the dimension of G .

Proof: K is the field of rationality of the isomorphism

$$H_{1,B}(A) \otimes_{\mathbf{Q}} \mathbf{C} \xrightarrow{\sim} H_{1,DR}(A) \otimes_k \mathbf{C}$$

given by “integration” over a cycle. This isomorphism preserves the cohomology classes of absolute Hodge cycles. Let P be the algebraic variety of isomorphisms having this property. From the definition of G and Theorem 1, it follows that P is a principal homogeneous space under G , defined over k , and therefore $\dim(P) = \dim(G)$. We conclude by the following lemma.

Lemma: Let z be a point of $\mathbb{A}^N(\mathbf{C})$; the transcendence degree of $k(z)$ over k is equal to the dimension of the k -Zariski closure of $\{z\}$;

This is a tautology.

Q.E.D.

Remark: Example 3 of Section 1 shows that $2\pi i \in K$. One can moreover show that the transcendence degree of K over $k(2\pi i)$ is bounded above by $\dim(G) - 1$.

Corollary: With the notation of Theorem 2, suppose in addition that A is simple of CM type. The transcendence degree of K over k is bounded above by $g + 1$.

Proof: It is enough to show that $\dim(G) < g + 1$. Now $\mu(\mathbf{G}_m)$ commutes with the action of the field E on $H_1(A, \mathbf{Q})$. It follows that G also commutes with this action and that G is an algebraic torus. Moreover, G preserves, up to a factor, the alternating form associated with a hyperplane section (see Example 3 of Section 1). This proves the assertion.

Observe that in this last case one has the following lower bound:

$$\dim(G) > 2 + \log_2(g),$$

due to Ribet [3].

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